Health & Safety

Standard Work with high pressure tools

Reference

Company	Document Type	Department	N°	Version	Langue
BYTP	REF	H&S	2224	A	EN

Version	Date	Write by	Checked by	Approved	Reason for modification
A	01-11-2023	O.BRZEZINSKI	I.LELIEVRE	L.KNOLL	First issue

The standards for the use of pressure equipment presented in this document serve as a practical guide to define the means to be planned and deployed to effectively ensure the completion of these activities.

Table of contents

1. PPE for work with high pressure

Specific high pressure PPE must be determined based on the effective working pressure.

500 Bar Gloves

Face shield and hearing protection

Combination 1000 bars in jet and 2800 in rotary

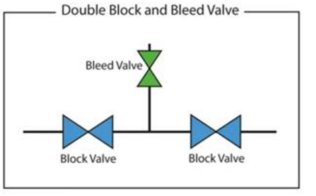
1,200 bar boots

2. Hydraulic networks

- Installation of block valves and purge valves according to the characteristics (pressures, temperatures, etc.) of the network
- Anti-whipping cables required
- Favor ball connections (easy to implement) with cat head fittings (quick connect system)
- Use of cable ties is prohibited

Anti-whipping cable

Identification of pipes



Example of double block and bleed system

Do not stay near a hydraulic or other network during a pressure rise phase for a test or in operation

3. Use of a karcher

Specific instructions related to the use of a karcher (non-exhaustive list):

- Risk assessment of the activity involving the task
- Specific operator training
- Equipment in perfect working order (hoses, lance, safety devices, etc.)
- Signage of the area
- On-site repair of high pressure equipment is prohibited
- Hold the lance firmly
- It is strickly forbidden to direct the jet towards another operator
- Wearing specific high pressure PPE
- **Never clean your boots with high pressure equipment**

Caution use of a high pressure jet

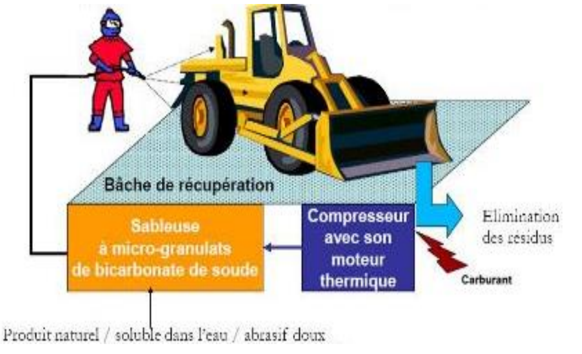
Karcher at pressure > 200 Bar = PROHIBITED
Contact the Prevention Department

4. Waterless washing of machinery

Advantages of the solution:

- Elimination of consumption - water and acid product discharges
- Ecological and quality cleaning of machinery
- Time saving in terms of stripping-cleaning operations, mobilization and repatriation of machinery compared to washing areas outside construction sites
- Reduced operating costs (labour, fuel and product)

Need for a covered area in rainy weather



5. Pressure washing of machinery

The main points of attention:

- Equipment compliant with local standards and regulations
- Washing water must be captured and treated
- The radius of curvature within the site must be taken into account so that vehicles can easily access the washing area

Washing platform

6. Compressed air networks

Installation principles:

- Air network equipped with valves (padlockable and accessible) allowing sections of the network to be isolated
- Air network equipped with purge valves
- **Use of cable ties prohibited**
- Compressor complies with local regulations and maintained in perfect operation
- Identifying Compressed Air Lines
- Visible pressure gauges
- Quickcoupling system fitted where applicable (prevent air from escaping when disconnected)
- Connection to fittings:
  - Possibility of installing steel cables inside pipes with a diameter > 2 inches instead of conventional anti-whipping cables
  - Antiwhip sling for all other diameters

Identification of pipes

Quick connectors

Diameter > 2 inches = inner steel cable

Anti-whipping cable

7. Others pressure equipment

Non-exhaustive list:

- Gas pressured equipment
- Vacuum equipment (depression)
- Steam pressure equipment

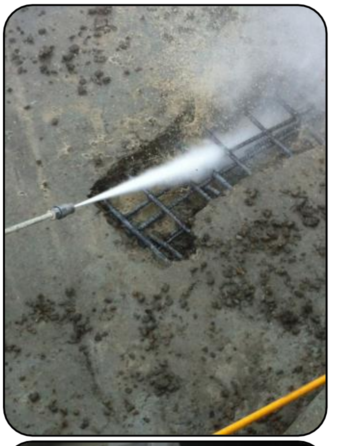
The installation, commissioning, use and maintenance of pressure equipment or networks are entrusted to competent personnel.

Example of pressure equipment label: gas cylinder

8. Hydrodemolition (subcontracting mandatory)

Hydrodemolition consists of the destruction of concrete using a high-pressure water jet (from 1,500 to 2,500 Bars), without causing potential harmful vibrations.
This technique allows selectivity of the areas to be treated, as only the “damaged” or fragile part will be eliminated.
The risk analysis will focus in particular on:

- Use of high pressure jet
- Material projections
- The noise generated
- The presence of running water
- The water mist/dust generated



No hydrodemolition nearby electrical installations

Exclusion zone: perimeter defined when carrying out the hydrodemolition, access prohibited to any person not participating in the task with danger display in place

Specific tarpaulins are necessary to protect neighbors (road traffic for example)

Wearing specific high pressure PPE is mandatory

9. Specific instructions in case of emergency

Anyone injured by a high-pressure jet must MANDATORY be examined by a doctor, regardless of the nature of the fluid, **including water**.

The injection of "non-sterile" water or other fluids under the skin has serious and irreversible consequences (infections, gangrene, amputation) if they are not treated effectively within the hours that follow.

In the event of violent shock due to a mechanical rupture of the pipe or hose, also consult a doctor (risk of internal bleeding).